

Level, Not State: Boundary Conditions for Phase-Structured Entropy in Long Reasoning Traces

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Abstract

We pre-registered a test of a popular idea: that long reasoning traces move through latent phases, and that a speculative decoder could read the phase to anticipate which drafted tokens survive verification. We ask whether a phase state inferred from the whole trajectory predicts acceptance beyond the instantaneous entropy level and a bandwidth-matched causal smoothing of it. The answer is null, and it holds at a ceiling: our state estimator sees each token’s future, so no deployable one can do better. Phase structure itself is real, but its scale is not identified. On the same 175 traces, with both estimators pre-registered and left at their defaults, a phase has a median length of 2 tokens under a hidden Markov model and 157 under changepoint detection, a factor of 78. The state adds nothing over the level because it is 93% a relabeling of it, a squashing function of the current token’s entropy rather than a slow regime. We report the boundary of the claim rather than its refutation, and name the analyst choices that quietly decide where it falls.

1 Introduction

Speculative decoding drafts tokens cheaply and verifies them in one pass, so its speedup rests on how many drafts survive (Leviathan et al., 2023; Chen et al., 2023). A widely held belief ties survival to entropy: where the drafter is uncertain, its guesses fail. A second idea sits beside it: long reasoning traces move through phases, exploring at high entropy and committing at low (Wang et al., 2025; Huang et al., 2025a; Xia et al., 2026). Together they suggest a decoder could find the phase, then draft further where it says acceptance is safe.

We pre-registered a test of that suggestion, fixing the fallback framing before we saw the data: does a phase state, inferable only from the whole

trajectory, predict acceptance beyond the instantaneous entropy level and a bandwidth-matched causal smoothing of it? Because our state estimator also reads each token’s future, it bounds any estimator a decoder could run, so a null at that ceiling is a null beneath it.

The answer is null. Phase structure is real, yet its timescale is not: two pre-registered estimators put its median length $78\times$ apart, and bimodality is rejected on 0.00% or 54.29% of them (section 4). The entropy correlation moves by a median factor of 4.57 under two definitions of acceptance (section 5). The state adds nothing over the level, correlating with it at 0.9333 (section 6). Our controller learns one draft length in every bin, and a constant policy recovers 101.5% of the oracle’s gain (section 7).

One shape runs through all four. The choice of null decided the bimodality verdict, the choice of estimator decided the timescale, the definition of acceptance decided the effect size, and an unstated assumption in a simulator decided which policy won. Each time the analyst’s choice fixed the headline number while the model’s behaviour did not move, and none reached print as a choice. We are not overturning this literature, and we did not find nothing. We report where the claim stops holding and name the mechanism that stops it: at this timescale, on this signal, the phase is a level.

2 Claims Under Test and Related Work

C1: long reasoning traces have phase structure. Established results show that high-entropy minority tokens drive effective reinforcement learning (Wang et al., 2025); PEAR shapes a reward around trace entropy phases (Huang et al., 2025a); EDRM reads reasoning as a dynamical system with entropy phase transitions (Xia et al., 2026). None concerns speculative decoding or claims phase predicts acceptance. Joining that structure to the

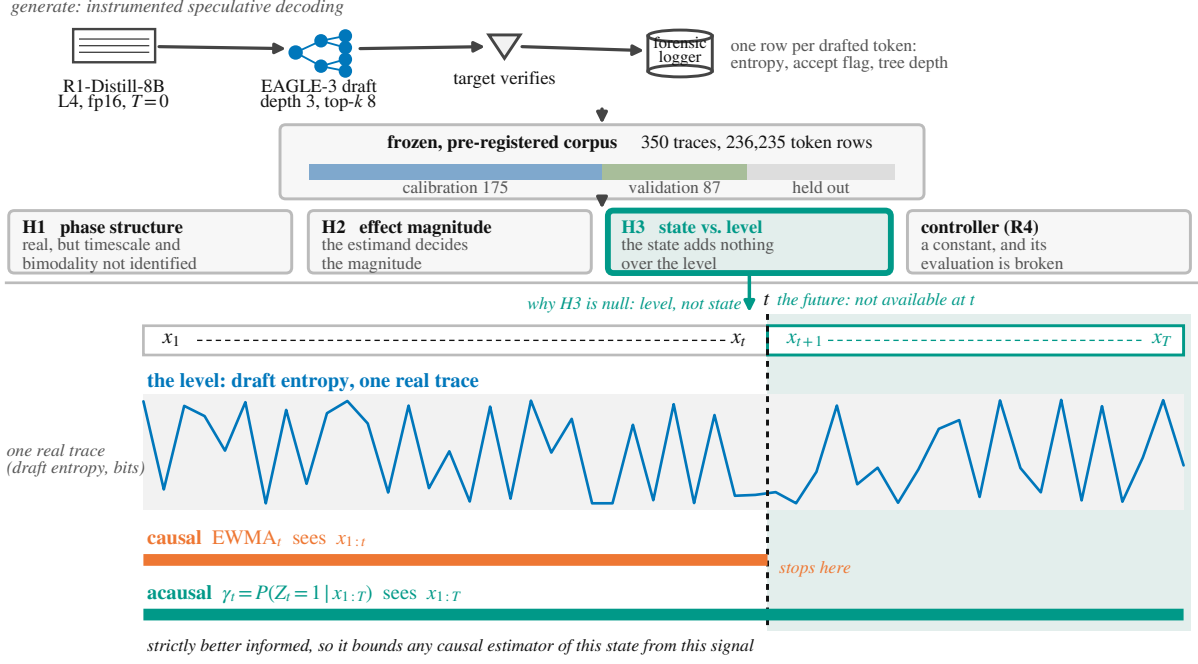


Figure 1: The study in one view. The top row is the pipeline: every drafted token is logged as the target verifies an EAGLE-3 proposal tree, producing a frozen, pre-registered corpus of 350 traces and 236,235 token rows that is split once into 175 calibration and 87 validation traces, with a third split held out. Four pre-registered analyses run on that corpus, and each returns a boundary condition rather than a phenomenon. The bottom row is why the state-versus-level verdict holds, read at the token level. The level (blue) is draft entropy from one real trace. The causal EWMA (orange) at token t has seen only $x_{1:t}$ and stops where the token does. The acausal HMM posterior $\gamma_t = P(Z_t=1 | x_{1:T})$ (teal) also sees the shaded future. Being strictly better informed, γ_t upper bounds any causal estimator of this state from this signal: if it adds nothing over the level, no deployable version can.

entropy link gives the deployable claim our pre-registration named: a latent phase carries information about acceptance beyond the entropy level at the token in hand. H3 tests it directly; H1 asks whether phase is identified at all.

C2: the reported magnitude of the entropy and acceptance correlation. SpecKV correlates draft entropy against acceptance and reports -0.549 , -0.559 and -0.567 across three quantisations in its Table 2 (Shukla, 2026). Acceptance Dynamics correlates target entropy against acceptance and reports -0.149 to -0.202 across four domains in its Table VI (Mahmoud, 2026). The two measure different entropy sources, and both agree with us on sign. SpecKV’s abstract quotes an unsigned 0.56, which is its confidence figure, not its entropy figure. Both are single-authored preprints without peer review: SpecKV runs twenty prompts and never names which correlation it computes; Acceptance Dynamics is five pages and generates sixty-four tokens per prompt. We therefore ask about the magnitude of a reported effect, not about a pillar of the field, and our an-

swer is not a failure to reproduce but a statement about the estimand (section 5).

Context. Speculative decoding drafts cheaply and verifies jointly (Leviathan et al., 2023; Chen et al., 2023). Our draft head comes from the EAGLE line (Li et al., 2024b,a, 2025). The controller we study belongs to the adaptive draft-length literature, which stops or extends a draft on a confidence or entropy signal (Agrawal et al., 2024; Mamou et al., 2024; Zhang et al., 2024; Huang et al., 2025b; Zhang et al., 2025; Yang et al., 2025) and reports end-to-end speedups. We ask the upstream question: does that signal carry the information the policy needs? A null there is informative rather than merely negative.

3 Corpus and Method

DeepSeek-R1-Distill-LLaMA-8B (Guo et al., 2025) is the target; EAGLE-3 (Li et al., 2025) the draft head. MATH-500 (Hendrycks et al., 2021; Lightman et al., 2024) and LiveCodeBench (Jain et al., 2025) generation uses `total_token=20`, `depth 3`, `top-k = 8`, `T = 0` greedy decoding,

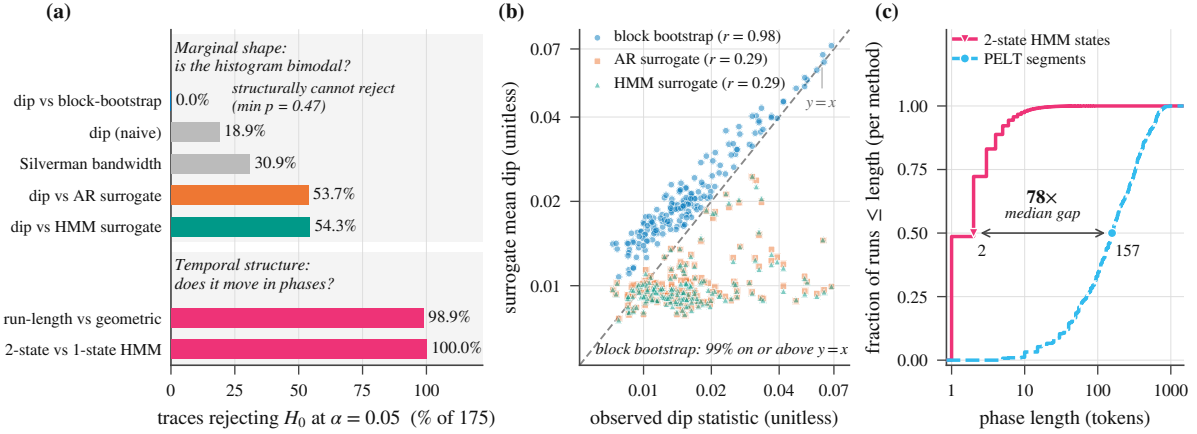


Figure 2: Phase structure is real, but neither its timescale nor its bimodality is identified. (a) Rejection rate by test and null. The temporal tests agree almost unanimously, while marginal bimodality rejection swings from 0.00% to 54.29% on the choice of null alone. (b) Observed dip statistic against each surrogate mean. The stationary block bootstrap sits on the $y = x$ line at $r = 0.9802$, because it resamples observed values and so preserves the marginal distribution it is supposed to be testing. (c) Phase length ECDF on a log axis. The same 175 traces give a median phase of 2 tokens under the HMM and 157 under PELT, a factor of 78.

fp16, single L4 GPU, and transformers 4.53.1 (Appendix B). Greedy decoding makes losslessness plausible, but the confirming audit could not run without controller-vanilla pairs, so Appendix H reports audit limitations, not a proof. Calibration has 175 traces (116,892 tokens), validation 87 (59,205 tokens), with a third held out and unused (Appendix C).

Signals recur: draft entropy, the current-softmax Shannon entropy; its bandwidth-matched causal EWMA_t with $\alpha = 0.4050$; and acausal $\gamma_t = P(Z_t = 1 \mid x_{1:T})$, the whole-trace forward-backward posterior of a two-state Gaussian HMM.

The contrast between EWMA_t and γ_t is our central design choice (Figure 1). EWMA_t is causal: at token t it has seen only tokens 1.. t . γ_t is acausal: computed from the whole trace, 1.. T , it has also seen everything after t , including how that passage eventually unfolds. Its α is matched to the HMM’s average phase persistence, so the comparison isolates causality rather than smoothing timescale. That asymmetry in information becomes a bound through the data-processing inequality: no causal, deployable phase estimator can carry more information about the hidden state at t than one that has already seen the entire trace. γ_t is therefore not one candidate detector among others; it is an upper bound on what any of them could achieve. A null result at that ceiling is decisive rather than a symptom of a weak estimator: we give the phase hypothesis an advantage no real system could have,

the unseen future, and ask whether it still buys anything over the level. If it does not, nothing causal built on the same entropy signal can do better.

The generator completes raw problem text without a prompt template, chat template, system prompt, or instruction wrapper. Traces therefore lack reasoning delimiters; the pre-registered delimiter-conditioned sub-analysis returned zero traces (Appendix I).

4 Phase Structure Is Real; Its Timescale and Its Bimodality Are Not Identified

Both pre-registered phase estimators ran over the same 175 calibration traces, from the same script, at package defaults, neither tuned. The two-state Gaussian HMM cuts them into runs of median length 2 tokens, 48.6% of them a single token; PELT (Killick et al., 2012), handed the same series, returns segments of median length 157. Two methods a reader would file under one heading disagree by 78 $\times$ on median phase length, against a mean trace length of 668 tokens (Figure 2(c)).

These estimators optimise different objectives, so their disagreement is a category error. That is correct, and it is the point: they share no estimand, so nothing in the data adjudicates. PELT’s penalty and the HMM’s state count are timescale knobs, so agreement can be manufactured on demand, and a quantity the analyst dials at will is one the data does not identify.

Two states beat one on 100.00% of traces and

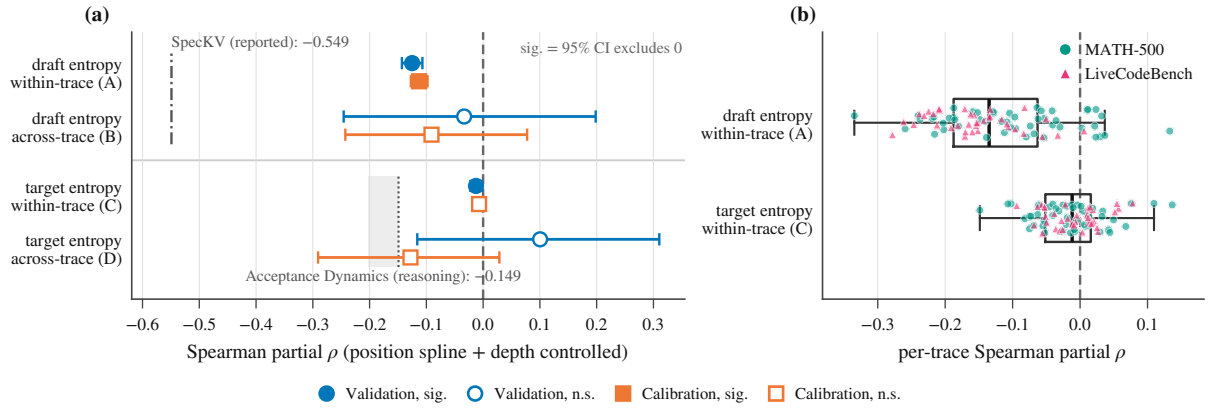


Figure 3: Each prior value belongs to only one half of the grid, and the cells that resolve sit well inside it. (a) Forest of the four cells on both splits. SpecKV measures draft entropy, so its -0.549 attaches to rows A and B; Acceptance Dynamics measures target entropy, so its band attaches to rows C and D. The gap is a property of the estimand rather than a failure to reproduce: correlating step-mean entropy against the step acceptance rate, as SpecKV does, moves cell A to -0.4357 on the same rows. (b) Per-trace correlations for cell A. MATH-500 (circles) and LiveCodeBench (triangles) separate, so the pooled estimate blends two per-domain effects.

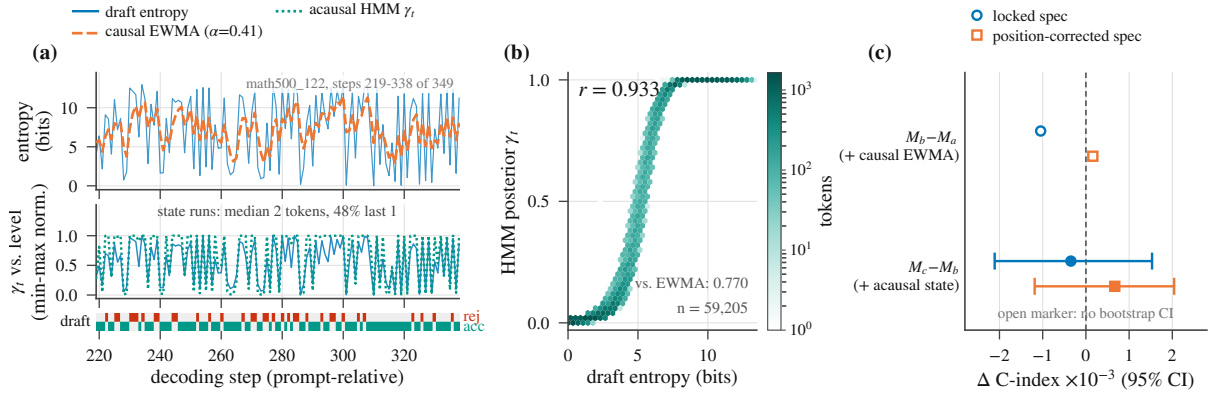


Figure 4: The state is a squashing function of the level, not a slow regime. (a) A windowed trace with the level, the causal EWMA and the acausal posterior overlaid. (b) γ_t against the level over all 59,205 validation tokens: a clean sigmoid at $r = 0.933$, against only 0.770 with the EWMA, so γ_t tracks the current token more tightly than any smoothing of it. (c) Incremental discrimination of the state over the level. Every interval crosses zero, under both position specifications.

the run-length test rejects a geometric law on 98.86%, but neither is calibrated for what we asked: the one-state model sits on the boundary of the two-state parameter space, where the chi-square reference fails, and the run-length statistic has a floor that geometric data clears just as easily. What survives is modest: a median effective sample size of 443.752 against a median of 697 tokens. The typical trace carries serial dependence, so entropy here is not white noise, but serial dependence is still not a phase.

Marginal bimodality is where the pre-registration breaks outright. With the same traces and dip statistic (Hartigan and Hartigan, 1985), rejection runs 0.00% against the stationary

block bootstrap, 53.71% against the AR surrogate, and 54.29% against the single-state HMM surrogate (Figure 2(a)). We do not read 53.71% as support for bimodality. Notice which pair agrees: the AR surrogate keeps the fitted autocorrelation while the single-state HMM surrogate is i.i.d. with only a mean and a variance, yet those two land together, and the block bootstrap, which keeps autocorrelation *and* the observed marginal, rejects nothing. The surrogate's marginal moves these verdicts, not its autocorrelation.

The block bootstrap fails hardest. It resamples observed values in blocks (Politis and White, 2004), and the dip is computed from the sorted sample, so block structure is inert: the null reduces

to an i.i.d. bootstrap of the observed marginal, the whole hypothesis. Its surrogate dip correlates $r = 0.9802$ with the observed dip it exists to challenge (Figure 2(b)), and per-trace p -values bottom out at 0.473 against the floor of 0.005 that 200 replicates make attainable, so no trace could have rejected at any effect size. Any resampling null that reuses observed values fails this way, and the block bootstrap is standard advice exactly when a series is autocorrelated. Appendix E carries the diagnostics, including the dip test’s clean bill against the other two nulls.

None of this is a fact about the model. The phase timescale and the bimodality verdict were fixed before anyone saw the data, and the rest of this paper watches that happen to an effect size and a metric.

5 The Estimand Decides the Magnitude

Reports that entropy predicts speculative acceptance leave two choices implicit: which distribution the entropy comes from (draft head or target), and what the correlation runs over (tokens within a trace, or whole traces). Crossing them gives a 2x2, estimated on the same tokens with a position spline and tree depth partialled out (Figure 3).

Only the within-trace cells resolve. Draft entropy against the logged acceptance flag gives $\rho = -0.1251$ (95% CI $[-0.1431, -0.1069]$) on validation and -0.1126 ($[-0.1273, -0.0979]$) on calibration. Target entropy gives -0.0126 ($[-0.0225, -0.0026]$), ten times smaller, on an interval disjoint from A’s. The across-trace cells are too wide to inform. Each prior belongs to half the grid: SpecKV (Shukla, 2026) measures draft entropy and reports -0.549 (A and B), Acceptance Dynamics (Mahmoud, 2026) target entropy and reports -0.149 to -0.202 (C and D). Everyone agrees on sign.

The $4.39\times$ gap to SpecKV reads as a failure to reproduce. It is not, and the evidence is internal. This project computed cell A under two analysis modes. One correlates step-mean draft entropy against the step acceptance rate, $\text{accept_length}/\gamma$ at $\gamma = 4$: SpecKV’s estimand. The other correlates per-token draft entropy against the binary logged accepted flag, the question a decoder faces, our headline. Six calibration traces went through both with identical row counts, so the same rows are read twice and only the estimand moves. Under SpecKV’s estimand cell A

is -0.4357 ($[-0.4962, -0.3709]$); under ours it is -0.1251 . Per trace the ratio runs $2.36\times$ to $18.25\times$, median $4.57\times$, and the two pooled intervals are disjoint. The six are representative under our own mode (mean -0.1015 against the full calibration -0.1126), so this is the estimand, not the sample. Adopting SpecKV’s estimand on our corpus lands within a factor of 1.26 of its reported value. The magnitude is a property of the estimand.

That comparison is a six-trace smoke test on calibration and cannot be rerun at scale: the H2 script and the raw corpus are outside our checkout. We report it as such.

The usual answer, that aggregation inflates correlations, is not true as stated. Aggregation inflates when the grouping moves both variables together, so between-group variance is signal, and attenuates when groups are exchangeable draws from one configuration, where it is noise. Traces are the second kind: cell B aggregates cell A’s tokens to trace means and $|\rho|$ falls to 0.0334, because the between-trace share of entropy variance is 0.0728, independently reproducing the project’s own ICC of 0.07138.

The shape recurs. A null choice swings H1 rejection from 0 to 54%, an unstated simulator assumption lets a constant policy beat the oracle, and here a $4.6\times$ factor rides on a choice between two defensible definitions of acceptance that papers report without reporting that a choice was made.

6 The State Adds Nothing Over the Level

Our confirmatory test fits a nested ladder of binomial GLMs (complementary log-log link, position spline, tree depth at acceptance): M_a (level) $< M_b$ (+ causal EWMA) $< M_c$ (+ acausal γ_t). The pre-registered per-trace random intercept was never fitted: the code uses `sm.GLM`, a plain GLM, leaving trace dependence to a cluster bootstrap over 87 traces. Appendix F records this deviation.

Across the locked run and both reconstructions, the pre-registered incremental discrimination (Harrell et al., 1982) of M_c over M_b is null: all three CIs cross zero (Table 1). The acausal state therefore adds no discrimination beyond the causal smoother. These are the locked run, a reconstruction differing from it through an unidentified model specification, and the prompt-relative reconstruction, which fixes the position bug and is primary. In each fit, adding the acausal state changes discrimination by less than one thousandth. This is

the same practical null under every available specification.

The likelihood-ratio test is an instrument, not evidence, because it is not reproducible from our artifacts. Position alone moves it from 37.97 to 40.73 (both approximately 10^{-10}); the five-order gap is between the locked 0.1767 and either reconstruction, not between position specifications. Both reconstruction JSONs carry `target_lr = 0.1767`, yet neither reproduces it. The test also reads 59,205 rows from 87 traces as independent, so our diagnostics trust the cluster-bootstrapped ΔC -index. The asymmetry runs against us: our primary specification is the one whose LR reads significant, so we are not cherry-picking. This makes the pre-registered test unavailable as reproducible evidence. A calibration gain would appear in the same likelihood channel, which is precisely the channel that fails to reproduce. Appendix D shows that 87 traces exceeded the 84 required for 80% power at a $+0.02$ absolute C-index effect.

Why does the state add nothing? Across 59,205 tokens, the sigmoid relation gives $\text{corr}(\text{draft_entropy}, \text{hmm_gamma}) = 0.9333$, $\text{corr}(\text{ewma}, \gamma) = 0.7696$, and $\text{corr}(\text{draft_entropy}, \text{ewma}) = 0.8102$. Thus the acausal state tracks the raw level more tightly than the level tracks its own causal smoother. The ordering, rather than each coefficient alone, identifies the mechanism. It is the sharpest version of the claim. Forward-backward smoothing had every chance to find a slow regime and returned a thermometer.

That collinearity is the result, not a flaw: the ΔC null and the 0.9333 are one finding seen twice, not two. It limits the ladder because testing whether a squashing of the level adds to the level approaches a tautology. Yet nothing pushed γ_t toward the level: the estimator ran freely over the whole trace and returned a phase with a median dwell of 2 tokens, nearly half its runs lasting one token. A label that changes every other token is not a regime.

Our title is correspondingly narrow. Because γ_t sees the whole trace, it bounds any causal estimator of the state it found. R1 locates that state at the two-state Gaussian HMM's 2-token median dwell; the 157-token PELT regime on the same traces is untested. The analyst picks this timescale, and the data does not identify it. On this signal, at this timescale, the phase is a level.

	locked	reconstruction	
	abs. pos.	abs. pos.	rel. step
M_a (level)	0.8503	0.8545	0.8410
M_b (+ EWMA)	0.8493	0.8545	0.8411
M_c (+ γ_t)	0.8489	0.8547	0.8418
ΔC ($M_c - M_b$)	-0.35	+0.15	+0.67
95% CI	[-2.11, +1.53]	[-1.48, +1.36]	[-1.19, +2.04]
LR M_c vs. M_b	0.18	37.97	40.73
p	0.674	7×10^{-10}	2×10^{-10}

Table 1: The H3 ladder across three fits. C-indices are on 59,205 held-out tokens from 87 traces; ΔC and its cluster-bootstrap CI (120 reps on `run_id`, `refit=True`) are in units of 10^{-3} . The prompt-relative reconstruction is primary. The effect size is null in all three fits, every CI crossing zero, while the pre-registered LR is recovered by neither reconstruction: the locked 0.1767 never reappears, and position indexing moves it only from 37.97 to 40.73. One statistic reproduces across fits and the other does not.

7 The controller is a constant, and its evaluation is broken

Can the level drive a deployable policy? Our controller reads the causal EWMA and learns a draft length per risk bin. At the frozen cost ratio ($c_{\text{fixed}} = 4$, $c_{\text{variable}} = 1$) it learns $[4, 4, 4, 4, 4, 4]$ across all six bins (`collapsed_to_single_length: true`), matching fixed $L=4$ to the last digit at 1.6672916138839624 tokens per call. The adaptive controller is a constant. The bins are not at fault: training cuts the risk score at its own quantiles and picks a length independently inside each, yet every bin chooses $L = 4$. The risk signal carries no actionable information for length selection.

The evaluation is the deeper problem. The project scores a policy by the recovered fraction $(\text{policy} - L_1)/(\text{oracle} - L_1)$ and reports 69.1% as an adaptation result. Apply it to the baselines, which never adapt (Table 2): fixed $L=8$ recovers 101.5%, a constant beating the oracle on its own scale.

Tokens per call is monotone in L only because the simulator scores lengths the corpus could never have produced. Generation ran at depth 3, so no verify call ever delivered more than 4 accepted tokens. Yet the simulator scores L as $\min(L, \text{safe_run})$, and `safe_run` counts contiguous accepts across a whole trace, reaching 147. Such a run is dozens of depth-3 calls credited as

policy	L per bin	tok/call	recov.	capped
fixed $L=1$	1,1,1,1,1,1	0.6627	(base)	(base)
fixed $L=2$	2,2,2,2,2,2	1.1184	31.3%	31.3%
controller	4,4,4,4,4,4	1.6673	69.1%	69.1%
fixed $L=4$	4,4,4,4,4,4	1.6673	69.1%	69.1%
fixed $L=6$	6,6,6,6,6,6	1.9659	89.6%	69.1%
fixed $L=8$	8,8,8,8,8,8	2.1392	101.5%	69.1%
oracle (HMM γ) acausal		2.1168	100%	

Table 2: Validation tokens per call and recovered fraction, $n = 59,205$ tokens. The controller’s row is identical to fixed $L=4$ ’s. Recovered fraction is (policy – fixed $L=1$) / (oracle – fixed $L=1$); the oracle is 100% by definition, and fixed $L=8$ exceeds it. The last column caps the simulator at the 4 accepted tokens the corpus’s depth-3 generation could actually deliver, which is where $L=6$ and $L=8$ lose their advantage. The controller’s 95% CI on recovered fraction is $[0.6781, 0.7065]$ (cluster bootstrap by run_id, 160 reps).

one draft accepted whole, never charged for compounding error. Lengths 6 and 8 therefore sit outside the corpus’s support, drawing 15.2% and 22.1% of their yield from runs longer than any real call. Cap the simulator at 4 and tokens per call saturates exactly at $L = 4$, both falling to 69.1%. The oracle runs on the same simulator, so its whole margin over the controller lives in the unsupported region.

The controller is a constant at the boundary of its corpus’s support, and $L=8$ was never a live option it missed. The project half-diagnosed this: its summary records the monotonicity, then repairs only the selection rule and leaves the pathology in the headline number. What decided that number was not the model’s behaviour but a simulator assumption nobody stated. Appendix G gives the objective and the $c_{\text{fixed}} \in \{2, 4\}$ sensitivity.

8 Discussion

Four results, one shape: the headline number moved because we changed the analysis, while the model, the corpus and tokens held still. Which null we compare against swings bimodality rejection from 0.00% to 54.29% on one set of traces, one statistic. Which estimator we call a phase detector swings the timescale by $78\times$: a median dwell of 2 tokens under the HMM, 157 under PELT, both at their pre-registered defaults. What we let acceptance mean swings the entropy correlation by $4.57\times$: on six traces read twice with identical row counts, step-mean entropy against a step

acceptance rate gives -0.4357 , the binary logged flag -0.1251 . What we call a state settles H3 before the ladder runs, since γ_t is 93% a relabeling of the level it was meant to beat. And one unstated simulator assumption lets a constant policy recover 101.5% of the oracle’s gain.

Each choice has a case for it, but papers report the number and not the choice, leaving the reader unable to separate the system from the analysis.

Report the estimand beside any entropy/acceptance correlation: what one row is, a token or verify step, and what acceptance means on it, a logged binary flag or $\text{accept_length}/\gamma$. A null is degenerate when its surrogate preserves the property the statistic reads. Correlate the surrogate’s statistic against the observed one: a null tracking the alternative at $r = 0.9802$ carries it inside. Then read the best p -value against the floor the replicate count allows: ours sat 95 times above it, an instrument that cannot move rather than a finding.

Limitations

Our corpus is one target model, DeepSeek-R1-Distill-LLaMA-8B, with one draft head, EAGLE-3, on two datasets, generated greedily at $T = 0$ with tree depth 3 and top- $k = 8$. Every number inherits that configuration and we do not know that any of it generalizes: a sampling decoder or a different draft head could move the entropy signal, and we ran neither.

Our 59,205 validation tokens are fewer observations than they look. The pilot’s ICC of 0.07138 and design effect of 29.86 sized the study for a $+0.02$ absolute C-index gain at 80% power, needing 1135 independent-equivalent rows and 84 traces; we ran 87. Our primary interval on ΔC runs $[-0.0011855764, +0.0020423226]$: it excludes the $+0.02$ we pre-committed to detect, and anything at or above roughly $+0.002$, but not a thousandth of a C-index. Absence of evidence for a state effect that small is not evidence of its absence. The pre-registered per-trace random intercept also went unfitted, so trace dependence rests on the cluster bootstrap instead.

The state we tested has a short scale: the HMM’s median phase runs 2 tokens, so our null concerns a 2-token state and nothing longer. PELT’s 157-token regime on these same traces is untested, since the original H3 fitting script is not in our checkout and we could not refit the ladder

at that scale. That is the honest limit of the title’s claim. Because γ_t correlates with the level at 0.9333, a reviewer can ask whether we tested a phase state at all, or only whether a near-copy of a variable adds to that variable. Section 6 takes that collinearity as the result rather than as damage to it, since nothing pushed γ_t toward the level. Still, if another estimator returns a state decorrelated from the level, our ladder says nothing about it.

Several of our own numbers can be quoted but not recomputed: the H2 and H3 fitting scripts sit outside our checkout and the raw corpus lives on a remote machine. Cell B reproduces exactly from the local features file at -0.0334 , since it needs only trace means. Cell A does not: its controls are absent from that file, and the local data returns -0.3899 against the published -0.1251 . The estimand comparison in Section 5 is limited the same way, a six-trace smoke test on calibration that cannot be re-run at scale. Its interval $[-0.4962, -0.3709]$ misses the full-corpus $[-0.1431, -0.1069]$ by a wide margin, but six traces are six traces.

Losslessness is unestablished rather than refuted: the audit gate reports FAIL on all five checks, and every one fails for a missing input rather than a detected violation. Separately, 961 of 236,235 reconstruction rows (0.407%) were flagged invalid with root cause “unresolved” and excluded rather than corrected.

Our own critique in Section 7 rests on a simulator the corpus cannot support, and that cuts against us too. We report the frozen evaluation’s numbers and the capped ones, but the cap is itself an assumption: it credits a length-8 draft with no more than depth-3 calls delivered, a floor rather than an estimate. We do not claim to know what a real $L = 8$ decoder would do, since drafting 8 deep compounds the draft head’s own error in a way no replay of this corpus can capture.

The deepest limit is our through-line. We pre-registered H1, H2, and H3, but not the claim that the analyst’s choices decide the headline number: that framing came together only after all four results returned null or degenerate. It is a hypothesis this study generated rather than one it tested, and no configuration of these four experiments would have falsified it. Our most quotable number carries the same defect in miniature, since both timescale estimators were pre-registered and the comparison between them was not. So we hold SpecKV and our own controller to a standard this paper does

not meet, the failure the paper is about. Four audits of one pipeline are also not four independent pieces of evidence about the field: we found every degenerate result in our own codebase, then generalized from it. Testing the claim means showing the same estimand, null, and metric sensitivity on a pipeline we did not write, and we have not done that.

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A Prompts, and What the Model Actually Saw

The corpus generator uses no prompt template. It reads a problem record from a dataset file, selects one text field out of that record, and hands the field to the tokenizer with nothing added around it. There is no instruction, no system message, no chat template, and no reasoning delimiter. The selection rule and the tokenizer call below are the whole of it, quoted from the generator.

No template: generate_corpus.py, lines 43-62 and 149-156

```
def sample_to_text(obj):
    if isinstance(obj, str):
        return obj
    if not isinstance(obj, dict):
        return json.dumps(obj)
    for key in ['question', 'problem', 'input', 'prompt', 'text']:
        if obj.get(key):
            return obj[key]
    if obj.get('question_content') or obj.get('question_title'):
        parts = []
        if obj.get('question_title'):
            parts.append(obj['question_title'])
        if obj.get('question_content'):
            parts.append(obj['question_content'])
        if obj.get('starter_code'):
            parts.append(obj['starter_code'])
```

```

        if obj.get('public_test_cases'):
            parts.append('public_test_cases: ' +
                          obj['public_test_cases'])
            return '\n\n'.join(parts).strip()
        return json.dumps(obj)

tokenized = model.tokenizer(
    text,
    return_tensors='pt',
    add_special_tokens=True,
    truncation=True,
    max_length=2048,
)

```

What fills it depends on which keys a record carries. A MATH-500 record stops the loop at its problem statement, so the model receives that statement and nothing else. A LiveCodeBench record carries no key from the loop’s list, so it falls through to the composite branch, and the model receives the four fields below joined by blank lines, with any absent field skipped. The literal string `public_test_cases:` is the only text the generator itself contributes to any prompt in this corpus.

Assembled LiveCodeBench text (generate_corpus.py, 51-61)

```

{question_title}

{question_content}

{starter_code}

public_test_cases: {public_test_cases}

```

A prompt builder that would have produced reasoning delimiters does exist in the repository, and it never ran against this corpus. It lives in the smoke-test harness, which generates single traces for debugging and writes no corpus. We reproduce it because the contrast is the point: had the corpus been generated through this path, the traces would carry an explicit opening delimiter and an instruction, and the pre-registered analysis that conditions on delimiters could have run.

The reasoning prompt that never ran (smoke_cot_trace.py, 22-40)

```

# dataset_name == 'math500':
{problem_text}

Please reason step by step, and put your final answer
within \boxed{ }.

# otherwise:
{problem_text}

Please reason step by step before giving the final answer.

# both are then wrapped and suffixed:
prompt = tokenizer.apply_chat_template(
    messages,                                # [{"role": "user",
    "content": user_text}]
    tokenize=False,
    add_generation_prompt=True,
)
return prompt + "<think>\n"

```

So the traces in this corpus are genuine DeepSeek-R1-Distill reasoning prose, produced because the model reasons by default rather than

setting	value	source line
total_token	20	generate_corpus.py:128
depth	3	generate_corpus.py:129
top_k	8	generate_corpus.py:130
use_eagle3	True	generate_corpus.py:125
temperature	0.0	generate_corpus.py:165
max_new_tokens	4096	generate_corpus.py:165
max_length	2048	generate_corpus.py:154
torch_dtype	float16	generate_corpus.py:133
GPU / precision	L4, fp16	implementation log
quantization	none	implementation log
transformers	4.53.1	implementation log

Table 3: Generation settings for the 350-trace corpus. The top block is read directly from the generator’s call to `EaModel.from_pretrained` and `eagenerate`. The bottom block appears only in the project’s implementation log, and no machine-readable field in the corpus records it.

because anything asked it to, and they carry no delimiters to condition on. Appendix J records what that cost the pre-registration.

B Generation Hyperparameters

Table 3 lists every generation setting with the line that fixes it. The tree parameters come from the generator’s own call and have not changed since the commit that introduced the file, which pre-dates the corpus by a day. We read them from the code rather than from any prose description, and Appendix J explains why that distinction matters here.

Two things in that table we could not verify from an artifact. The GPU and the library version appear in the implementation log and nowhere else: the corpus files, the split locks, and the frozen evaluation JSON record no hardware identifier and no library version, so a reader has to take the log’s word for both. The pinned transformers version sits in the log’s dependency list, which covers the repository as a whole rather than the corpus run in particular. Prompt truncation never binds in practice, since the generator warns when an input reaches 2048 tokens and the corpus contains no such record, but the cap is part of the configuration and we report it.

Every trace was generated twice, once with logging off and once with logging on, and only the logged pass produced the corpus. The target is DeepSeek-R1-Distill-LLaMA-8B (Guo et al., 2025) and the drafter is the matching EAGLE-3 head (Li et al., 2025), verifying a token tree in one pass. The datasets are MATH-500 (Hendrycks

et al., 2021; Lightman et al., 2024) and LiveCodeBench (Jain et al., 2025), at 200 and 150 problems.

C Forensic Logger Schema

The logger writes one JSON record per drafted token, and the corpus holds 236,235 such records across 350 traces, stored as Parquet partitioned by model, drafter, dataset, and temperature. Each record identifies its trace through `run_id` and its source problem through `problem_id`, and places the token in the sequence through `step`, which counts from the end of the prompt, and position. The token itself appears as `token_id` and `token_str`.

Four fields carry the distributions this paper measures. `draft_entropy` and `target_entropy` are Shannon entropies in bits over the drafter’s and the target’s full softmax, `draft_top1_prob` is the drafter’s top-1 probability, and `draft_topk_probs` holds the drafter’s top 32 probabilities. Acceptance and tree behaviour occupy four more: `accepted` is the verifier’s boolean outcome for the token, `tree_depth_at_accept` and its synonym `tree_depth` record the depth at which the token was resolved, and `accept_length` records the cumulative accepted run. Two fields are reserved for analysis rather than written at generation: `phase_label_hmm` is null in the frozen corpus and gets filled when the HMM is fit, and `is_inside_think` marks membership of a delimited reasoning span. The remainder are auxiliary: `best_candidate`, `error`, and four tensor-shape strings kept for debugging. Analyses drop rows where error is non-null.

Two of those fields deserve a warning. `is_inside_think` exists and never fires, because the generator emits no delimiters, so the field is false throughout and every analysis keyed on it returns nothing (Appendix J). Separately, the schema document states that `tree_depth_at_accept` never exceeds 3, and the project’s own corpus-level summary reports a mean above that bound, so the two disagree. We therefore treat the generator’s arguments, not the schema’s prose, as the authority on the tree configuration.

The schema also decides what the losslessness audit could and could not do. No field records tree candidate identity, node parentage, or whether a

node lay on the selected path, so a row cannot be mapped back to the verified path it came from. Appendix I returns to that.

The corpus carries a `split` field, null at generation and assigned afterwards by grouping on `problem_id` so that no problem crosses a split. The calibration lock holds 100 MATH-500 and 75 LiveCodeBench problems, giving 175 traces and 116,892 token rows, and the validation lock holds 50 and 37, giving 87 traces and 59,205 rows. The two share no problem. Everything this paper reports is fit on calibration and evaluated on validation, and the remaining problems stay held out and unused. We audited the third lock file and found it corrupt, which changes nothing here because no reported number touches it; Appendix J gives the detail and the reason we describe the test split as unused rather than as clean.

D Power Analysis

The pre-registration sized the corpus for H3 and fixed its target before any data existed: detect an increment of 0.02 in absolute C-index at 80% power and $\alpha = 0.05$. The pilot supplied the dependence structure. Across 50 pilot traces the intraclass correlation of the per-token signal was 0.07138, and the mean trace ran 405.28 token rows.

Those two numbers are what make 59,205 tokens worth far less than 59,205 tokens. Tokens inside one trace are not independent draws: they share a problem, a prompt, and a decoding trajectory, so each new token from a trace already underway repeats much of what its neighbours said. The standard correction prices that repetition as a design effect, $1 + (\bar{k} - 1) \cdot \text{ICC}$, which at $\bar{k} = 405.28$ and $\text{ICC} = 0.07138$ gives 29.86. A correlation of 0.07 sounds negligible, and it is not, because the design effect multiplies it by the trace length: hundreds of weakly correlated rows compound into a factor of thirty. One token row therefore carries about a thirtieth of the information an independent row would, and the effective sample size lands nearer the trace count than the row count.

The sizing follows. Reaching 0.02 at 80% power needs 1,135 independent-equivalent rows, which at a design effect of 29.86 across traces of 405.28 rows means 84 traces. We reproduce that arithmetic exactly from the pilot figures. The validation split holds 87 traces, so the confirmatory test ran at the size it was designed for, and the

test	reference	reject
<i>marginal shape</i>		
Hartigan dip	tabulated	18.86%
Silverman bw test	bootstrap	30.86%
GMM, 2 vs. 1	BIC	98.86%
dip	block bootstrap	0.00%
dip	AR(p) surrogate	53.71%
dip	1-state HMM surr.	54.29%
<i>temporal structure</i>		
HMM, 2 vs. 1 states	χ^2	100.00%
run-length KS	geometric	98.86%

Table 4: Every H1 test over the same 175 calibration traces, same script, package defaults, $\alpha = 0.05$. The dip statistic is identical across the three surrogate rows; only the reference distribution changes, and the verdict travels from never to most traces.

interval we report in the body is about ten times narrower than the effect the design was powered to find. The same logic disciplines everything else in the paper: any test that treats the 59,205 rows as independent, including the likelihood-ratio tests in Table 1 and the Fisher z in Appendix J, is reading a sample size roughly thirty times larger than the one it has.

E H1: Full Per-Null Results and Why the Block Bootstrap Cannot Test Bimodality

Table 4 gives every H1 test at its pre-registered settings over the 175 calibration traces, at $\alpha = 0.05$. The temporal tests agree with each other and the marginal tests do not agree with anything.

The three surrogate rows share a statistic and differ only in what they hold fixed, which is where the pre-registration broke. The AR(p) surrogate fits an autoregression and draws fresh Gaussian innovations, so it keeps the trace’s autocorrelation. The single-state HMM surrogate fits a one-state Gaussian model and samples from it, which has no transition structure at all, so its draws are i.i.d. and it keeps only a mean and a variance. Those two hold opposite things about time and land 53.71% against 54.29%, agreeing on 96% of traces. The block bootstrap keeps autocorrelation and the observed marginal together, and rejects nothing. Time is not what moves these verdicts. We note this because the script’s own docstring calls all three nulls autocorrelation-preserving, which is true of two of them.

The block bootstrap’s failure is structural rather than statistical. It resamples observed values in

null	r with obs. dip	p range
block bootstrap	0.9802	[0.473, 1.000]
AR(p) surrogate	0.2913	[0.005, 0.970]
1-state HMM surr.	0.2851	[0.005, 0.950]

Table 5: Why the block bootstrap cannot reject. Its surrogate dip tracks the observed dip almost exactly and exceeds it on 98.9% of traces at a mean ratio of 1.331, so its smallest p -value over 175 traces is 0.473 against an attainable floor of 0.005. The other two nulls reach the floor.

blocks (Politis and White, 2004), so each surrogate is the original numbers in a new order with some drawn twice. The dip statistic (Hartigan and Hartigan, 1985) is computed from the sorted sample and never sees order, so block structure is inert for it and the stationary bootstrap reduces to an i.i.d. bootstrap of the observed marginal. The script concedes as much: when the sampler raises, its fallback substitutes plain resampling with replacement, which for this statistic is the same null. The surrogate therefore reproduces the marginal whose shape is the entire hypothesis under test, and null and alternative become the same distribution.

The diagnostics in Table 5 confirm it and give the direction of the bias. Across the 175 traces the block bootstrap’s mean surrogate dip correlates $r = 0.9802$ with the very statistic it exists to challenge, and sits above it: the mean ratio of surrogate to observed is 1.331, and 98.9% of surrogate means land at or above the observed value. Resampling with replacement adds ties and gaps to the empirical distribution, and ties and gaps inflate a dip on their own, so this null is biased upward of the thing it tests. The per-trace p -values follow with no exception: the smallest across all 175 traces is 0.473 and the median is 0.930. Two hundred replicates put the attainable floor at $(0 + 1)/(200 + 1) = 0.004975$, so the best trace in the corpus sits 95 times above the smallest p -value the test could ever return. No trace could have rejected, at any effect size, and a pipeline that pre-registered this null alone would have shipped 0.00% across 175 traces as a confirmed negative. The other two surrogates generate new values instead of recycling old ones, and their correlations with the observed dip drop to $r = 0.2913$ and $r = 0.2851$, which is what a null that varies the quantity under test looks like.

One further H1 view is worth recording because it returned nothing. The pre-registration named

three views: the raw trace, a formatting-masked trace, and the longest contiguous delimited reasoning span. The first two ran on all 175 traces. The third returned `n_traces: 0` and wrote an empty file, for the reason Appendix A gives.

F H2: Full Grid, Estimand Comparison, and Ecological Ladder

The full H2 grid in Table 6 restores the cells and the calibration split omitted from the body. The within-trace intervals resolve on validation, while both across-trace intervals cross zero. Calibration preserves the strong draft-entropy result in cell A, but its other three intervals cross zero.

The estimand comparison holds the traces and row counts fixed while changing the definition of acceptance. The `paper_step_spearman` mode correlates step-mean draft entropy with the five-level step acceptance rate `accept_length/γ` at $\gamma = 4$. This is SpecKV’s estimand. It contrasts with `h2_reconciliation_partial`, which correlates per-token draft entropy with the binary logged accepted flag. Table 7 gives both pooled estimates and the representativeness check.

Table 8 shows the matched comparison in full. The absolute ratio spans the second and sixth rows, and the summary row gives its median. The matched token mean lies close to the full calibration estimate in Table 7, so the subset represents that mode rather than selecting an unusually weak sample. This remains a six-trace smoke test on calibration. We cannot rerun it at scale because the H2 script and the raw corpus are outside the checkout.

The ecological ladder isolates a second choice. It holds the variables and the Spearman estimator fixed on the same 59,205 validation tokens, then changes only the grouping. Table 9 includes the ungrouped baseline and every grouping in the artifact. Its trace-mean rung equals published cell B at -0.03340745060873369 , including the first ten decimal places. That exact reproduction licenses the remaining rungs derived from the same local feature file.

The ladder supplies the condition missing from the usual aggregation claim. Aggregation inflates a correlation when the grouping moves both variables together, making between-group variance signal. It attenuates when groups are exchangeable draws from one configuration, making between-group variance noise. Our traces

are the second kind. Their between-trace entropy variance is 0.9154759461931022 , while within-trace entropy variance is 11.663052661652534 , so the between-trace share is 0.07278084541796806 . This independently reproduces the project’s own ICC of 0.07138 and explains why trace means erase rather than concentrate the token-level association.

The local feature file does not reproduce published cell A. Its uncontrolled within-trace rung is -0.3898664677602963 , against the published -0.1250526137859289 . Cell A requires tree depth at acceptance and possibly other controls absent from that file, and its fitting script is not in the checkout, so we can quote the published cell but cannot recompute it. Cell B does not need those controls and reproduces exactly from trace means.

The full validation summary also records a naive Fisher test comparing cells A and C, $z = -19.466165636386208$, that treats all 59,205 clustered tokens as independent. At the project’s effective sample size of 84, the same calculation gives about $z = -0.72$, $p = 0.47$; we do not use it and rely on the disjoint cluster-bootstrap intervals instead.

G H3: Specification, Collinearity, and the LR Clustering Analysis

Each rung of the ladder is a binomial GLM with a complementary log-log link over one row per drafted token at risk, with rejection as the event. Every rung carries a B-spline in position with $df = 5$ and the tree depth at acceptance. M_a adds the instantaneous draft entropy, M_b adds the causal EWMA at the matched $\alpha = 0.40502282239836346$, and M_c adds the acausal posterior γ_t . We fit on the 175 calibration traces and evaluate on the 87 validation traces, and the increment $M_c - M_b$ carries a cluster bootstrap over `run_id` with 120 resamples and `refit=True`.

Three departures from the pre-registered specification matter here. The pre-registered M_a named draft confidence and an AdaEDL bound term (Agrawal et al., 2024) alongside draft entropy; the fitted M_a carries draft entropy, tree depth, and the spline, so two named predictors never entered the ladder. The pre-registered per-trace random intercept was never fitted, and trace dependence reaches the result only through the cluster bootstrap, which is why the ΔC -index is the statistic we trust and the likelihood-ratio test is not. The

split	cell	estimate	CI lower	CI upper	<i>n</i>	<i>k</i>
validation	A	-0.1250526137859289	-0.14307675744176007	-0.10694554794858138	59,205	87
validation	B	-0.03340745060873369	-0.2456469778807576	0.1983134382461349	59,205	87
validation	C	-0.012567123424001512	-0.02249577930851252	-0.0026359888404600762	59,205	87
validation	D	0.10038638186192315	-0.1159883308448829	0.31034563026932904	59,205	87
calibration	A	-0.1126270789246881	-0.1273148774068361	-0.09788989622331462	116,892	175
calibration	B	-0.09104572942552334	-0.24265113854898435	0.07751770750868252	116,892	175
calibration	C	-0.007089036032805461	-0.014309043388669087	0.00013171051759203477	116,892	175
calibration	D	-0.12804866102143803	-0.2906556383140195	0.028604806643256882	116,892	175

Table 6: The complete H2 grid on both locked splits. A correlates draft entropy with acceptance within trace, B correlates their trace means, C substitutes target entropy within trace, and D correlates the target-entropy trace means. Each interval comes from the reported cluster bootstrap, and *k* counts traces.

mode and sample	unit	ρ	CI lower	CI upper	<i>k</i>
step, matched calibration	verify step	-0.43566574775672645	-0.49617140758148687	-0.370948875143514	6
token, full validation	token	-0.1250526137859289	-0.14307675744176007	-0.10694554794858138	87
token, full calibration	token	-0.1126270789246881	-0.1273148774068361	-0.09788989622331462	175
token, matched calibration	token	-0.10152164995822187	mean over matched traces		6

Table 7: The pooled estimand comparison and representativeness check. Step is paper_step_spearman, using step-mean entropy and accept_length/ γ at $\gamma = 4$. Token is h2_reconciliation_partial, using token entropy and the binary logged acceptance flag. The matched token row is an arithmetic mean, so it has no pooled interval.

locked run also indexed the spline on absolute sequence position rather than prompt-relative step, which we corrected.

Table 10 reports all three fits in full. The conclusion is the same in each: the increment is a few ten-thousandths of a C-index and its interval crosses zero. The likelihood-ratio test is the opposite. Neither reconstruction recovers the locked LR = 0.1767, and both reconstruction JSONs carry that value as an unmet target. Position indexing moves the statistic only from 37.97 to 40.73, so the five orders of magnitude between the locked run and the reconstructions belong to a specification difference the artifacts could not identify, not to position.

The collinearity is severe and it is the finding rather than an obstacle to it. Over the 59,205 validation tokens, γ_t correlates with draft entropy at 0.9333332475 and with the EWMA at 0.7696206508, while draft entropy correlates with its own EWMA at 0.8102448436. A posterior computed from the whole trace therefore sits closer to the current token than a causal smoother of that token does. The design matrix condition number for M_c is 172.8523992916 in the corrected fit and 201.3939477690 in the locked one, which is high enough for the locked run’s own diagnostics to raise a high-correlation warning and low enough that the fits are not numerically degenerate.

The clustering analysis deserves care, because it is easy to over-read and its own summary over-

reads it. What it establishes is that the likelihood-ratio test is computed unclustered across 59,205 rows drawn from 87 traces, and that the cluster-bootstrapped ΔC -index is the metric to trust instead. What it does not establish is that clustering explains the discrepancy. Its own cluster-adjusted permutation test, permuting γ_t within each of the 87 traces, returns the observed LR = 37.97 against a permutation mean of 5.62 and a 95th percentile of 14.15, for an empirical *p* of 0.0 and a recorded still_mismatches_target: true. The clustering it blames is weak on its own numbers: the intraclass correlation of tree depth at acceptance is 0.029, which its narrative nonetheless calls highly within-cluster correlated. Its within-trace correlation between tree depth and γ_t averages -0.4126. The two artifacts also disagree about how many permutations ran, one saying 500 and the other 1000, for the same statistic and the same result.

The same analysis tried three alternative placements of tree depth, and the result is the strongest evidence that the locked run is gone. Putting tree depth in M_a only reproduces the locked likelihood ratio almost exactly, at 0.0805 with *p* = 0.7766, and collapses the M_b and M_c C-indices to 0.7205. Putting it in all three models reproduces the C-indices to within 0.006 and leaves the likelihood ratio at 37.97. Dropping it entirely reproduces the likelihood ratio and misses the C-indices by more than 0.13. No tested specification reproduces both, so the locked configuration is not recoverable from

trace	step-rate ρ	token-flag ρ	rows in each mode	absolute ratio
livecodebench_0	-0.4141399255494481	-0.0462128962800992	631	8.96156611867217
livecodebench_101	-0.5136855117548692	-0.2174295135872906	674	2.3625381084644785
livecodebench_104	-0.4600853677762262	-0.1251298678707711	640	3.6768628913712527
math500_1	-0.5095941674924748	-0.13991282698249	852	3.6422262238776018
math500_10	-0.3262247789884995	-0.0596846686784154	962	5.465805310007136
math500_101	-0.378948477024582	-0.0207601263502649	746	18.253669107353318
mean, with ratio median	-0.4337797047643499	-0.10152164995822187	4,505	4.571334100689194

Table 8: The same six calibration traces under the step acceptance-rate and binary token-acceptance estimands. Each trace contributes the identical row count to both modes, and the last column is the absolute ratio of the two correlations. The final row gives the arithmetic mean for each correlation, the total matched rows, and the median ratio.

grouping	ρ	recorded count
pooled token rows, no trace control	-0.3658177100490384	59,205 rows
token rows within trace, no controls	-0.3898664677602963	87 trace groups
trace means	-0.03340745060873369	87 groups
dataset by step octile	-0.2529411764705882	16 groups
step-octile means	-0.8095238095238096	8 groups
dataset by global step 16-tile	-0.27602639296187675	32 groups
global step 16-tile means	-0.8294117647058824	16 groups
entropy 16-tile means	-1.0	16 groups

Table 9: The ecological ladder on one set of validation tokens. The variables and rank-correlation estimator stay fixed while the grouping changes. The count is the number of token rows for the pooled baseline and the number of groups for every other rung. Dataset by step octile supplies the SpecKV-like grouping, and entropy bins expose the ceiling when grouping aligns with both variables.

what we hold, and we report the ladder under the specification that matches the C-indices while declining to use its likelihood ratio as evidence.

H The Controller Objective, Its Sensitivity, and the Limits of the Simulator

The controller learns one draft length per risk bin. Training cuts the risk score at its own quantiles into six bins, digitizes the score against those edges, and selects a length independently inside each bin by maximizing an efficiency proxy,

$$\text{efficiency}(L) = \frac{\text{tokens_per_call}(L)}{c_{\text{fixed}} + c_{\text{variable}} \cdot L},$$

with the frozen ratio $c_{\text{fixed}} = 4$ and $c_{\text{variable}} = 1$. The project documents that ratio as a placeholder standing in for verification overhead until measured hardware costs replace it, and it is the reason the selection rule has an interior optimum: the numerator grows with L and saturates, the denominator grows without saturating, so efficiency rises to a maximum at $L = 4$ and falls at 6 and 8. Every bin picks $L = 4$ because that optimum does not move with risk. The bins themselves are sound, and we say so because the natural diagnosis is

wrong: the digitizing step partitions the tokens correctly and the selection runs per bin.

What is not sound is the diagnostic block that reports the bins. The frozen JSON records counts of 116,892 in all six bins, which is the entire calibration split six times over, and six identical mean_values of 0.19723498746561072. Neither is a binning failure. The reporting code re-derives per-bin membership after the fact by matching each token’s assigned length value rather than by reusing the bin identifiers, so when the policy collapses to a single length every token matches every bin, and each bin reports the whole split and the split’s grand mean. The degeneracy is exactly co-extensive with the collapse it should have exposed, which is what makes it worth naming.

The sensitivity that actually exists covers two values of c_{fixed} . At the frozen $c_{\text{fixed}} = 4$ the policy is $[4, 4, 4, 4, 4, 4]$. At $c_{\text{fixed}} = 2$ a second configuration is constructed and fitted, and it returns $[4, 2, 2, 2, 2, 2]$, reaching 1.2206232581707626 tokens per call and a recovered fraction of 0.3836655515286683. Halving the fixed cost moves the optimum toward shorter drafts and the policy stops being constant, which is the shape Figure 5(b) plots. No third value was ever run; Appendix J explains what the project re-

	locked	reconstruction	
	abs. position	abs. position	rel. step
M_a log-likelihood	-15275.995	-15456.278	-15502.852
M_b log-likelihood	-15259.877	-15445.660	-15492.323
M_c log-likelihood	-15259.788	-15426.676	-15471.958
M_a AIC	30571.990	30928.555	31021.703
M_b AIC	30541.753	30909.320	31002.646
M_c AIC	30543.577	30873.351	30963.916
M_a C-index	0.8503221418	0.8544698455	0.8409574994
M_b C-index	0.8492750572	0.8545143725	0.8411176217
M_c C-index	0.8489274259	0.8546645150	0.8417854887
$\Delta C (M_c - M_b), \times 10^{-3}$	-0.3476313	+0.1501425	+0.6678671
CI lower, $\times 10^{-3}$	-2.1113223	-1.4837148	-1.1855764
CI upper, $\times 10^{-3}$	+1.5309244	+1.3562018	+2.0423226
LR M_b vs. M_a	32.2367914526	21.2354565591	21.0572190921
p	1.36e-8	4.06e-6	4.46e-6
LR M_c vs. M_b	0.1766644513	37.9686060692	40.7300220747
p	0.674	7.19e-10	1.75e-10
cond. number (M_c)	201.3939477690	107.1846123946	172.8523992916

Table 10: The H3 ladder across all three fits, evaluated on 59,205 validation tokens from 87 traces. The ΔC -index interval crosses zero in every fit, which is the pre-registered outcome and the paper’s conclusion. The likelihood-ratio test does not survive re-fitting at all: the locked 0.1767 reappears in neither reconstruction, and both reconstructions record it as a target they failed to match.

ported instead.

The simulator scores lengths the corpus never produced

The reductio in the body has a root cause, and it changes what the monotonicity means. The corpus was generated at depth = 3, so a single verify call could deliver at most three drafted tokens plus one bonus token from the target, and the step-level table confirms the ceiling: accept_length never exceeds 4.0 in any step of any trace, with one row per step.

The simulator does not respect that ceiling. It scores candidate lengths $\{1, 2, 4, 6, 8\}$ against safe_run, which counts contiguous accepted tokens forward from each position across the whole trace, with no regard for which verify call produced them. On the validation split safe_run reaches a maximum of 147 and exceeds 4 on 17.03% of tokens. A run of 147 accepted tokens is roughly thirty-seven separate depth-3 draft calls, each of which received a fresh target-verified context. Treating it as evidence that one draft of length 147 would have been accepted whole ignores what makes long drafts hard: a drafter proposing eight tokens deep conditions on its own predictions and compounds its own error, and nothing in the corpus measures that.

Lengths 6 and 8 therefore sit outside the cor-

L	sim. tok/call	capped	from safe_run > 4
1	0.6627	0.6627	0.0%
2	1.1184	1.1184	0.0%
4	1.6673	1.6673	0.0%
6	1.9659	1.6673	15.2%
8	2.1392	1.6673	22.1%

Table 11: The simulator’s advantage for long drafts is extrapolation. Capping safe_run at 4, the largest accepted run any single verify call in this corpus ever produced, saturates tokens per call at $L = 4$ and equalizes the recovered fraction of $L = 4$, $L = 6$, and $L = 8$ at 69.1%. The last column is the share of each length’s simulated yield drawn from accepted runs longer than any real call delivered.

pus’s support, and their advantage is extrapolation. Table 11 caps the simulator at the 4-token maximum a real call actually delivered and re-runs it. Tokens per call saturates exactly at $L = 4$, the recovered fraction for $L = 6$ and $L = 8$ falls to the same 69.1% that $L = 4$ earns, and the 101.5% reductio disappears. Our capped recomputation reproduces the frozen baselines to ten decimal places at $L \in \{1, 2, 4\}$, which is where the corpus can speak.

So tokens per call is not intrinsically monotone in L . It is monotone because the simulator extrapolates past the corpus’s support with no depth penalty, and the hindsight oracle is built on the same simulator and inherits the same flaw, which

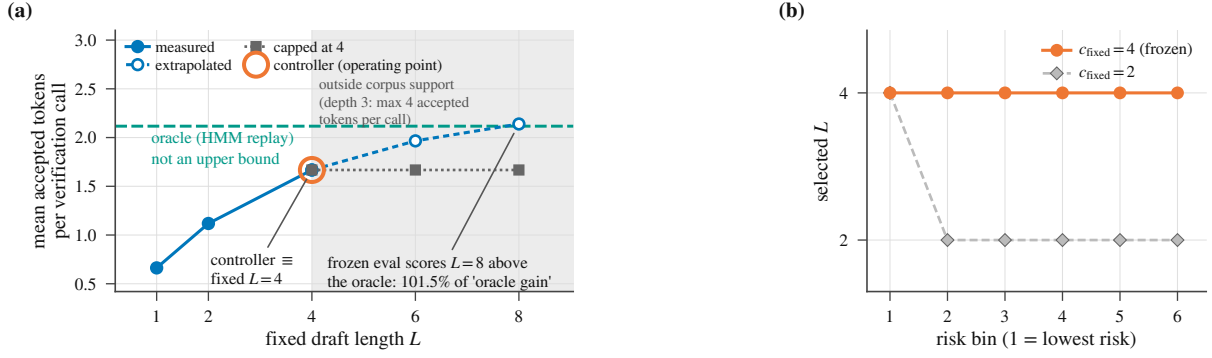


Figure 5: The controller at the frozen cost ratio. (a) Mean accepted tokens per verification call against fixed draft length L . The solid stretch is measured; past $L=4$ the curve is drawn open and dashed because the corpus, generated at tree depth 3, never delivered more than 4 accepted tokens in one call, so the shaded region is outside its support. The controller’s operating point (ring) sits exactly on the $L=4$ dot, and the frozen evaluation scores fixed $L=8$ above the oracle line. The capped curve restricts the simulator to the 4 tokens a call could actually deliver and goes flat from $L=4$. (b) Selected length per risk bin: flat at $c_{\text{fixed}} = 4$, stepped at $c_{\text{fixed}} = 2$. The policy adapts only away from the frozen operating point.

is how a constant policy comes to beat it. This sharpens the body’s claim rather than softening it: the controller is a constant sitting at the boundary of what its training corpus can describe. A constant is still not adaptive, and fixed $L = 4$ still reproduces it to the last digit, but choosing $L = 8$ was never a live option the controller missed.

The cap cuts against us as well, and we report both columns rather than substituting one for the other. Crediting a length-8 draft with no more than a depth-3 call ever delivered is a floor, not an estimate: a real drafter running 8 deep might do better than the cap allows and would certainly do worse than the uncapped simulator claims, because it would be conditioning on its own predictions the whole way down. The corpus cannot settle that in either direction. What it does settle is that the frozen evaluation’s ranking of $L = 6$ and $L = 8$ above $L = 4$ rests on an assumption the corpus never tested and the project never stated.

I What the Audit Establishes, and What It Does Not

This paper does not prove that speculative decoding was lossless here, and the audit that would have proven it did not run. The losslessness gate reports an overall status of FAIL on all five checks, and Table 12 gives the reason recorded for each. Every one is a missing input rather than a detected violation. Nothing in the audit found a token that differed from what the target would have produced alone; the audit never got far enough to look.

The honest position is therefore narrow. Greedy

check	recorded reason
sampler_invariant	(mis-specified at $T=0$)
tost_distribution	controller/vanilla pair
_equivalence	not provided
holm_fwer_positions	no per-position TOST p -values
greedy_token	missing controller or
_identity_t0	vanilla outputs
eqspec_ragged	missing batch audit
_desync_batch_gt_1	JSONL

Table 12: The losslessness gate, overall status FAIL. Each check failed for want of an input, so losslessness is not established and not refuted.

decoding at $T = 0$ makes losslessness plausible on the argument the project itself records, that tree verification under argmax-prefix acceptance returns the target’s own greedy sequence, and the controller changes only how many tokens are proposed rather than the acceptance criterion. That is an argument, not a measurement, and the project’s status file states it as proven while the empirical gate beside it fails on every check. We report the gap rather than the conclusion. Two things would close it: a controller and vanilla output pair from the same prompts, and the instrumentation named in Appendix C, since the forensic rows carry no candidate or path identity and a row cannot be traced to the verified path it belonged to.

One more limitation belongs here. The P4.1 reconstruction re-derived draft probabilities for all 236,235 rows and flagged 961 of them, 0.407%, as invalid, because the reconstructed probability exceeded the live logged top-1 probability beyond

tolerance. The largest such gap is 0.7451496, which is not a rounding artifact. The report gives the root cause as unresolved, naming execution-path numerical drift and context reconstruction mismatch as competing explanations without deciding between them, and the flagged rows are excluded from P4.1 statistics rather than corrected. The rate is close to uniform across datasets, at 0.409967% for LiveCodeBench and 0.404462% for MATH-500, which is consistent with a systematic reconstruction issue rather than a property of either corpus. No result in this paper depends on those rows, and we have not explained them.

J Deviations from the Pre-Registration

This is the appendix we would most want to read in someone else’s paper. The pre-registration named both outcomes publishable before we ran anything, which is what makes the null worth reporting; it also means every gap between the plan and the execution belongs on the record. We group them by kind rather than by severity.

The analysis specification changed in three ways. The locked H3 run indexed the position spline on absolute sequence position where prompt-relative step belonged. We found the error, corrected it, made the corrected fit primary, and report both. The null holds under both, and under the third fit as well: the increment is -0.0003476313 , $+0.0001501425$, and $+0.0006678671$ across the three, and every interval crosses zero, so the headline does not depend on which side of the bug a reader stands. Separately, the pre-registered M_a named draft confidence and an AdaEDL bound term that never entered the fitted ladder, and the pre-registered per-trace random intercept was never fitted, leaving the cluster bootstrap to carry trace dependence. Appendix G gives all three.

The controller is not the detector we pre-registered. The plan specified an EWMA-hysteresis detector with two thresholds, an upper one to enter the decision phase and a lower one to leave it, calibrated by grid search to maximize predicted acceptance. The controller we built has no hysteresis and no thresholds. It bins a risk score at its own quantiles and picks a length per bin by an efficiency proxy. The pre-registered rejection budget of 0.12 turned out to be infeasible on the real corpus and collapsed the policy to length 1 every-

where, and the efficiency proxy replaced it. The project documents that substitution openly. We report it here because the detector that reached the paper is not the one the pre-registration describes.

One sub-analysis could never have run. The corpus was generated with no chat template, so the traces carry no reasoning delimiters, and the pre-registered H1 view restricted to within-delimiter spans returned `n_traces`: `0` and wrote an empty file. This was not a null result. There was nothing to analyze, and the field the analysis keys on, `is_inside_think`, is false for every row in the corpus. The pre-registration assumed a generation path the generator never took, and no check caught it, because a view returning zero traces looks like a filter that excluded everything rather than a design error.

Two reported numbers have no artifact behind them. The first is a live sweep. The project’s results summary reports a wall-clock comparison over 30 LiveCodeBench problems, with mean times for an autoregressive baseline, a fixed-length baseline, and the controller. Those numbers appear in the summary’s prose and nowhere else. No file in `results/` contains a wall-clock or throughput field, no P4.2 artifact exists, and the last commit touching `results/` precedes the commit that integrated the controller into the decode path by about three hours on the same day. We dropped the sweep. This paper reports no wall-clock number of any kind, and its controller results are offline replays on a frozen corpus.

The second is a sensitivity analysis. The frozen evaluation JSON carries a `policy.sensitivity_checked` field reporting selected lengths at $c_{\text{fixed}} \in \{2, 4, 8\}$, and the project’s policy document and results summary both quote it. It is a hardcoded literal. The controller script writes those three lists into the results dictionary as constants, and no configuration with $c_{\text{fixed}} = 8$ is constructed anywhere in the codebase, so no such policy was ever fitted. Serialized into the JSON beside computed fields, the literal is indistinguishable from a result. The tell is internal: the literal’s own $c_{\text{fixed}} = 2$ entry reads `[4, 4, 4, 2, 2, 2]`, while the configuration actually constructed and fitted twenty lines later returns `[4, 2, 2, 2, 2, 2]`, and the results summary quotes the fabricated list rather than the computed one. Only $c_{\text{fixed}} \in \{2, 4\}$ exist. We report the computed values in Appendix H and none of the

literal.

Several published numbers can be quoted but not recomputed. The H2 and H3 fitting scripts are not in our checkout, and the raw corpus lives on a remote machine, so the derived files are all we hold. That is enough for some numbers and not others. Cell B reproduces exactly from the local token features, at -0.03340745060873369 to every digit, because it depends only on trace means and those match the H2 corpus. Cell A does not. The local file gives -0.3898664677602963 against the published -0.1250526137859289 on the same traces with the same row counts, and no lag and no combination of local columns reproduces the published value. The published triples are internally consistent, so H2 computed a real Spearman; we simply cannot recompute which one. The project’s own summary reports an unpartialled supplemental cell A identical to the partialled headline, which means the controls alone do not explain the gap either. This is a limitation of our artifact set rather than a defect in the result, and it is the same theme the body reports: a named quantity is not a quantity until the estimand is named with it.

The corpus configuration is recorded twice, inconsistently. The generator hardcodes `total_token = 20`, `depth = 3`, and `top_k = 8`, unchanged since the commit that introduced the file, and the corpus schema corroborates the depth and the `top-k`. The implementation log records 60, 5, and 10. Those belong to a different run, the Spec-Bench reproduction on a T4 in 8-bit, which the log describes two lines above them and which produced no analysis data. Appendix B reports what the generator ran. We flag it because the log is the document a reader would reach for first, and because a results file and an implementation log disagreeing about the corpus configuration is exactly the kind of drift a reproducibility appendix exists to catch.

The test split lock is corrupt, and no reported number touches it. The calibration and validation locks are clean and consistent with everything we report: 100 MATH-500 and 75 LiveCodeBench problems for calibration, giving 175 traces, and 50 and 37 for validation, giving 87, with no overlap between them. The test lock is not. It holds 125 MATH-500 problems and zero LiveCodeBench problems, where the corpus split

implies 88 test traces drawn from both datasets. Worse, 58 of its ids also appear in the calibration lock and 29 in the validation lock, so only 38 of its 125 entries are held out of training at all. This violates the project’s own leakage-prevention lock, and the file cannot be what produced the 88-trace test split the corpus documents. It changes nothing in this paper because no reported number touches the test split: every confirmatory result runs on calibration and validation, and the controller is fitted on calibration and evaluated on validation. The accurate statement is that calibration and validation are locked, and the test split is held out and unused. It should not be described as a clean 88-trace held-out split, and we do not describe it that way.

Two smaller things, for completeness. The H1 script’s `docstring` calls its three nulls autocorrelation-preserving, and one of them is not: the single-state HMM surrogate draws i.i.d. Gaussian values and preserves a mean and a variance only. The label is wrong in the script, and Appendix E states what each null holds fixed. Finally, the H2 summary’s pairwise test separates cells A and C with a Fisher z computed as though all 59,205 tokens were independent, which is the assumption this paper spends Appendix D pricing. Fisher z scales with the square root of the sample size, so recomputing it at the 84 effective units the project’s own power analysis reports shrinks it by more than an order of magnitude and leaves it far from any conventional threshold. No claim in this paper uses it; the cluster bootstrap intervals for the two cells are disjoint without it.